

Data Quality vs. Quantity

The LIMA result: 1,000 carefully curated examples rival 52,000 — why quality dominates

9 min

Duration

B4 – Analyse

Bloom's

L2 – Intermediate

Level

AI-FT-IN-TH-000014

ID

The LIMA Finding — Less Is More for Alignment

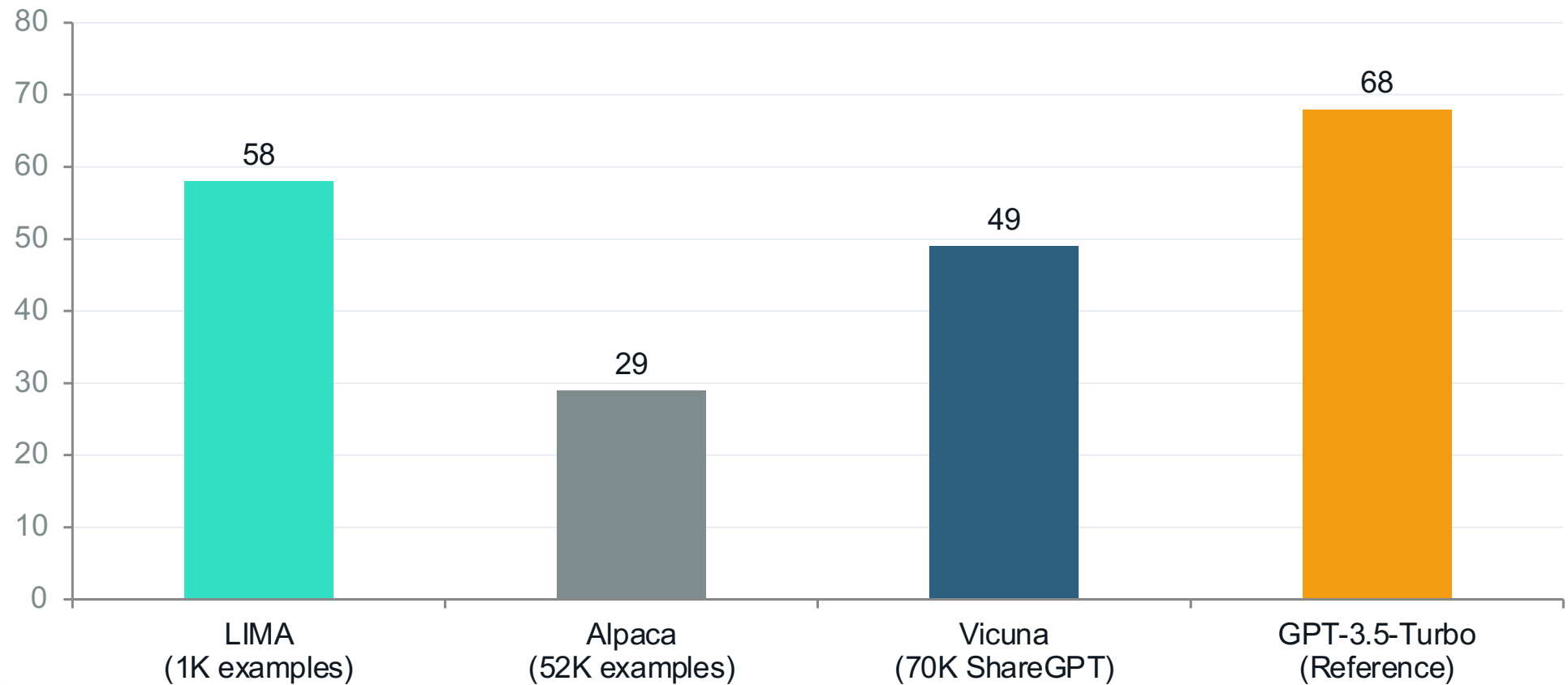
LIMA (Zhou et al., NeurIPS 2023)

A model fine-tuned on just 1,000 carefully selected, diverse, high-quality examples (LIMA dataset) matches or outperforms models fine-tuned on 52,000 GPT-generated Alpaca examples — on human preference evaluations.

- Interpretation: almost all of the knowledge needed for instruction following is already present in the pre-trained model; fine-tuning teaches the model the surface format to express it
- Alignment Tax: large noisy datasets can actually hurt performance on reasoning tasks by flooding the model with low-quality patterns
- Practical implication: invest heavily in dataset curation, not just volume

LIMA vs Alpaca — Human Preference Results

Human Preference Win Rate — LIMA vs Baselines



Data Selection Strategies — Applied to Practice

Perplexity-Based Filtering

Remove examples where the base model already performs well (low perplexity). Focus training signal on hard cases the model doesn't yet know.

Diversity Sampling

Use k-means clustering on instruction embeddings. Sample proportionally from each cluster to ensure balanced coverage.

LLM-as-Judge Scoring

Use GPT-4 or Claude to score each (instruction, response) pair on a 1–5 rubric. Keep only examples scoring ≥ 4 .

Deduplication First

Always run MinHash deduplication before any other filtering. Near-duplicate examples waste compute and introduce repetition bias.

KEY REFERENCES

- Zhou, C. et al. (2023). LIMA: Less Is More for Alignment. NeurIPS 2023. arxiv.org/abs/2305.11206
- Chen, B. et al. (2023). Alpargusus: Training a Better Alpaca with Fewer Data. arXiv.

 **Next: Concept 18: Building Instruction Datasets (AI-FT-IN-CD-000004)**